(12) FILE SYSTEM = COMPOSITE PATTERN

→ Composite pattern :- structural pattern used when group of object should be treated as single type object

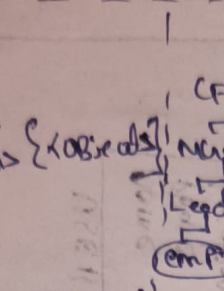
→ Problem : objects will be in tree structure form

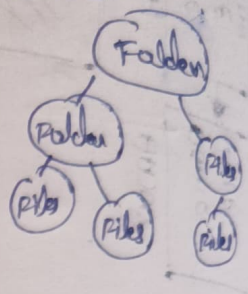
File{ name }

Folder{ name List<File> List<Folder> {<Object>} }

for(Object:obj){

for(object item: fileFolder){
if(item instanceof File){
((File)item).show()
else if
}}





→ Every time we should use instance of & do casting
→ So many if else conditions.

SOLUTION

(Interface)
<FileSystem>
show();

1 → Many

File
name
show()

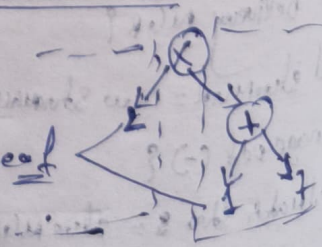
FOLDER
List<FileSystem>
name
show()

→ Here we don't need to use the if, else & type casting.
→ Because both file & folder implement common FILESYSTEM.
⇒ both class obj has same Parent.

for(FILESYSTEM item: ft){
item.show()
}

ARITHMETIC EXP

2 × (1+7)



① Leaf nodes are numbers.
② we have Left | Right.

